

Module Reference & Review

All 154 modules: what each one does, the technique it actually uses, and its verified runtime status.

An engineering review of all **154 modules** in Perimetry v2.0: what each one does, the technique it actually uses, and its verified runtime status. Every status in this document comes from a real execution run, not from reading code.

How this was verified. All 154 modules were executed as subprocesses against a live target (`example.com` , the IANA-reserved documentation domain) with a 60-second per-module cap, plus a static pass over all 188 Python files for syntax, imports and catalog integrity.

1. At a glance

METRIC	RESULT
Modules in catalog	154
Executed cleanly (exit 0)	143 / 154
Python tracebacks across the whole run	0
Long-running (exceeded the 60 s test cap)	8
Correctly rejecting wrong input type	3
Require an API key for full output	18
Average module size	145 lines
Sections	Network & Infrastructure · Web Application Analysis · Security & Threat Intelligence

Headline: the suite is in good working order. Zero modules crash. The only non-zero exits are three modules correctly refusing a domain when they need a CIDR or ASN, and eight modules that are simply slow because they do a lot of network work.

2. How a module works

Every module follows the same contract, which is what makes the suite consistent:

```
python -m perimetry.modules.<name> <target> <threads> [json-options]
```

1. **The runner spawns it as its own process** (`perimetry/core/runner.py`). One module crashing can never take down the CLI, and each gets a clean interpreter.
2. `run(target, threads, opts)` is the entry point. `opts` is the JSON blob of per-module options set with `set <id> <option> <value>` .
3. **Output is printed with** `rich` - tables and panels - and captured by the runner line by line, so progress is visible while the module is still working.
4. **Results are written twice:** a per-module `.txt` under `results/<domain>/` , and a combined report for the whole run.
5. **Severity is inferred** from the text the module prints, which the runner maps to OK / WARN / ALERT for the summary.

The practical consequence: a module is just a script that prints. That is why the suite scales to 154 of them without central coordination - and also why output quality varies, since nothing enforces a common result schema.

3. The modules that matter most

Not all 154 modules carry equal weight. These are the ones that would define Perimetry as a serious security tool.

Tier S - the flagship ten

These combine a genuinely useful finding, real implementation depth, and no dependency on a paid API.

ID	MODULE	WHY IT IS TOP-TIER
154	JARM TLS Fingerprint	The single most technically demanding module. Byte-exact with Salesforce's reference implementation - verified identical on 6 independent hosts - so its hashes can be looked up directly in public JARM/C2 datasets. Pure sockets, zero dependencies.
152	JS Secret / API Key Scanner	Highest direct bug-bounty value in the suite. 22 vendor credential patterns plus a Shannon-entropy filter, severity grading, and masked output so the report never leaks the key it found.
156	Subdomain Takeover (Deep)	Turns a critical, immediately actionable finding. 20-service fingerprint database, dual CT sources (crt.sh + CertSpotter) so one provider outage does not blind it, and marker confirmation to suppress false positives.

ID	MODULE	WHY IT IS TOP-TIER
19	Zone Transfer	The classic critical DNS misconfiguration. An open AXFR hands an attacker the entire internal zone; still found in the wild.
116	Git Repository Exposure Check	An exposed <code>.git</code> directory means full source-code recovery, often with credentials in history. Cheap check, severe finding.
96	Exposed Environment Files Checker	A served <code>.env</code> is game over - database passwords and API keys in plaintext.
113	Cloud Bucket Exposure	Permutes bucket names across S3, GCS and Azure. Public-bucket data leaks remain one of the most common real breaches.
87	DOM Sink Scanner	Genuine client-side vulnerability analysis - traces DOM XSS sinks (<code>innerHTML</code> , <code>eval</code> , <code>document.write</code>) back to their sources, rather than just listing headers.
151	TLS Security Config	The deepest TLS audit in the suite: protocol versions, cipher strength, forward secrecy and known protocol weaknesses in one verdict.
153	Security Headers Grade	Converts seven weighted controls into a single A+ -to- F grade. The kind of one-number output that makes a tool feel professional and gets read by non-specialists.

Tier A - strong supporting modules

ID	MODULE	STRENGTH
158	Document Metadata Extractor	Classic OSINT that most scanners skip. Handles both the PDF info dictionary and modern XMP packets, and recovers real employee names for phishing-risk assessment.
155	CVE Enrichment from Tech Stack	Turns passive fingerprints into ranked, CVSS-scored CVEs via the NVD API. Honest about being a keyword match.
157	Subresource Integrity Checker	Does not merely check that an <code>integrity</code> attribute exists - it re-downloads each asset and recomputes the hash, catching stale hashes that silently break production.
90	CSP Deep Analyzer	Parses the policy into directives and grades real weaknesses (<code>unsafe-inline</code> , wildcards, missing <code>default-src</code>).
148	Email Config	The most complete email-posture module: MX, SPF, DKIM, DMARC plus MTA-STS and BIMI.
4	DNSSEC Check	Walks the actual DNSKEY/DS/RRSIG chain rather than just reporting presence.
111	CT Log Query	Certificate transparency is the single richest passive source for subdomain discovery.
112	Breached Credentials Lookup	Uses the HIBP k-anonymity range API, so no complete hash ever leaves the host - a genuinely privacy-correct implementation.
68	Favicon Hashing	Small but strategically valuable: a favicon hash pivots to every other host running the same application.
34	RPKI Route Validity Check	Advanced networking most recon suites lack - detects BGP hijack exposure.
43	SNMP Public Community Checker	Default-credential exposure that still yields full device inventories.

ID	MODULE	STRENGTH
63	CORS Misconfiguration Scanner	Replays forged <code>Origin</code> headers to find genuinely exploitable trust.
85	GraphQL Introspection Probe	A readable schema hands an attacker the entire API surface.
130	Attack Surface Delta	Async subdomain + resolution + port pipeline; the closest thing to continuous monitoring.

The dependable core

Unglamorous but used in nearly every engagement, and all verified working: **3** DNS Records · **5** Domain Info · **18** WHOIS Lookup · **99** HTTP Headers · **13** SSL Expiry Alert · **27** CDN Detection · **60** Technology Stack Detection · **67** Form Grabber · **104** Security.txt Check · **118** SPF/DKIM/DMARC Validator.

Honest assessment - the weakest links

Quantity is not the same as depth. These are the thinnest modules and the first places to invest:

ID	MODULE	ISSUE
149	Dark Web Monitoring	49 lines - the smallest module. Depends on <code>psbdmp.ws</code> and <code>scylla.sh</code> , both frequently offline. The name promises far more than it delivers.
108	Subdomain Enumeration	62 lines, single source (<code>crt.sh</code>). Module 156 supersedes it with multi-source enumeration.
102	Pastebin Monitoring	Produced 15 characters of output - effectively nothing without a GitHub token.
51 / 52 / 57	Crawler, Robots.txt Analyzer, Redirect Chain	Thin wrappers around a single request each; reasonable as utilities, not as findings.
128	Custom Wordlist Generator	A helper for other tools rather than a recon module in its own right.

4. Full module reference

Legend:  pass ·  long-running (correct, just slow) ·  needs a different input type.

4.1 Network & Infrastructure (54 modules)

ID	MODULE	WHAT IT DOES & HOW IT WORKS	KEYS	STATUS
1	Associated Hosts	Reverse host lookup to list domains sharing an IP. Queries HackerTarget, Shodan and crt.sh for hostnames sharing the target's IP, then merges and de-duplicates them.	Shodan	✅ pass
2	DNS Over HTTPS	Resolve DNS records via encrypted DoH endpoints. Sends DNS-over-HTTPS queries to Cloudflare, Google, Quad9 and AdGuard in parallel and compares the answers.	—	✅ pass
3	DNS Records	Enumerate standard DNS RRsets (A, AAAA, MX, NS, etc.). Resolves A/AAAA/MX/NS/TXT/CNAME/SOA record sets concurrently with dnspython.	—	✅ pass
4	DNSSEC Check	Detect and validate DNSSEC configuration. Walks the DNSSEC chain - DNSKEY, DS and RRSIG records - and reports whether the zone is signed and validates.	—	✅ pass
5	Domain Info	Registrar, creation/expiry, and zone metadata. Combines WHOIS, RDAP and DNS to report registrar, creation/expiry dates and zone metadata.	—	✅ pass
6	Domain Reputation Check	Aggregate trustworthiness indicators from reputation sources. Checks the domain against VirusTotal, AlienVault OTX and Cisco Talos reputation feeds.	Otx, Virustotal	✅ pass
7	HTTP/2 and HTTP/3 Support Checker	Detect server support for HTTP/2 and HTTP/3 (QUIC). Negotiates ALPN over TLS for h2 and opens a QUIC connection via aioquic to confirm HTTP/3.	—	✅ pass
8	IP Info	Geo, ASN, and ownership info for target IPs. Geolocates and enriches the resolved IP through ip-api.	—	✅ pass
9	Open Ports Scan	TCP port scan to identify exposed services. TCP-connect scans common ports with a thread pool, optionally enriched with Shodan's port data.	Shodan	🕒 long-running
10	Server Info	Gather server banners, stack hints, and versions. Reads the HTTP response headers and resolved IP to describe the serving stack.	—	✅ pass

ID	MODULE	WHAT IT DOES & HOW IT WORKS	KEYS	STATUS
11	Server Location	Approximate server geolocation & hosting provider. Resolves the host and looks its IP up in a geolocation API.	—	✓ pass
12	SSL Chain Analysis	Retrieve cert chain; validate trust path & intermediates. Opens a TLS socket and walks the presented certificate chain, printing issuer/subject per link.	—	✓ pass
13	SSL Expiry Alert	Check certificate expiration window; warn when near expiry. Fetches the leaf certificate and computes days remaining until notAfter, flagging near expiry.	—	✓ pass
14	TLS Cipher Suites	Enumerate supported TLS cipher suites. Repeatedly handshakes offering one cipher family at a time to enumerate what the server accepts.	—	✓ pass
15	TLS Handshake Simulation	Simulate varied TLS client handshakes; flag issues. Simulates handshakes across TLS versions and reports which succeed plus the negotiated parameters.	—	✓ pass
16	Traceroute	Trace network hops to the destination. Runs a hop-by-hop trace toward the target and renders the path.	—	✓ pass
17	TXT Records	Retrieve TXT records (SPF, DKIM, verification tokens). Pulls and pretty-prints the TXT record set (SPF, verification tokens, etc.).	—	✓ pass
18	WHOIS Lookup	WHOIS/RDAP ownership data retrieval. Shells out to the system whois client and parses the registrar response.	—	✓ pass
19	Zone Transfer	Attempt AXFR to enumerate full DNS zone when misconfigured. Attempts an AXFR zone transfer against each authoritative nameserver - a classic misconfiguration check.	—	✓ pass
20	ASN Lookup	Map IPs/domains to ASNs & network orgs. Maps the IP to its ASN and announcing organisation via RDAP/ip-api.	—	✓ pass

ID	MODULE	WHAT IT DOES & HOW IT WORKS	KEYS	STATUS
21	Reverse IP Lookup	Enumerate domains hosted on a given IP. Reverse-lookups the IP through Shodan and HackerTarget to list co-hosted domains.	Shodan	✓ pass
22	IP Range Scanner	Scan an IP range for live hosts & open ports. Expands a CIDR with the ipaddress module and probes each host in the range concurrently.	—	⚙️ needs CIDR
23	RDAP Lookup	Structured domain/IP ownership via RDAP. Queries the RDAP registry (the structured successor to WHOIS) for domain or IP objects.	—	✓ pass
24	NTP Information Leak Checker	Query NTP servers for version & info leak data. Sends NTP mode-6/7 control queries to detect monlist-style information leaks and amplification exposure.	—	✓ pass
25	IPv6 Reachability Test	Validate IPv6 DNS + connection reachability vs IPv4. Tests IPv6 reachability across an address range, requiring IPv6 CIDR input.	—	⚙️ needs IPv6 CIDR
26	BGP Route Analysis	Inspect BGP announcements & paths for anomalies. Pulls prefix, peer and upstream data for the target's ASN from BGPView.	—	✓ pass
27	CDN Detection	Detect CDN fronting (Cloudflare, Akamai, etc.). Correlates CNAME chains, response headers and known CDN IP ranges to name the CDN in front of the site.	—	✓ pass
28	Reverse DNS Scan	PTR sweeping to discover hostnames. Performs reverse-DNS (PTR) lookups across a netblock with a thread pool.	—	✓ pass
29	Network Timezone Detection	Approximate timezone from geo/latency/banner clues. Infers the server's timezone by comparing its clock/geolocation against worldtimeapi.	—	✓ pass
30	Geo-DNS Footprint	Compare DNS answers across global resolvers; map geo/ASN variance. Resolves the domain from several vantage resolvers to reveal geo-steered DNS answers.	—	✓ pass

ID	MODULE	WHAT IT DOES & HOW IT WORKS	KEYS	STATUS
31	SPF Network Extractor	Parse SPF includes/mx/a; expand & extract sending netblocks. Recursively expands the SPF record's include/ip4/ip6 mechanisms into the full sending netblock set.	—	✓ pass
32	NS Geo/ASN Diversity Analyzer	Assess authoritative NS geo & ASN concentration. Geolocates every nameserver and scores how diverse their countries and ASNs are (resilience check).	—	✓ pass
33	DNS SLA Latency Monitor	Measure resolver latency & SLA metrics; flag slow responders. Times repeated resolutions against each authoritative nameserver to build a latency/SLA picture.	—	✓ pass
34	RPKI Route Validity Check	Validate route origins for target prefixes against RPKI VRPs. Checks whether the announcing ASN is RPKI-valid for the prefix using RIPEstat - BGP hijack exposure.	—	✓ pass
35	Recursive Nameserver Leak Test	Detect recursion enabled on authoritative nameservers. Sends recursion-desired queries to the target's own nameservers to detect open recursion.	—	✓ pass
36	Dual-Stack Behavior Profiler	Compare HTTP/TLS responses over IPv4 vs IPv6; flag diffs. Compares behaviour over IPv4 vs IPv6 - resolution, TLS and response - to find dual-stack drift.	—	✓ pass
37	ICMP Reachability Matrix	Ping sweep; build loss/latency matrix; detect filtering. Builds an ICMP reachability matrix across the resolved addresses.	—	✓ pass
38	IP Allocation History Tracker	Historical IP allocation & ownership timeline. Reconstructs the IP's allocation and transfer history from RIPEstat.	—	✓ pass
39	Autonomous Neighbor Peering Map	Map upstream/downstream AS adjacencies. Maps the ASN's BGP peers and neighbours into a peering graph via BGPView.	—	✓ pass
40	TLS Session Resumption Map	Probe TLS session resumption across hosts. Tests TLS session ticket and session-ID resumption support across the endpoint set.	—	✓ pass

ID	MODULE	WHAT IT DOES & HOW IT WORKS	KEYS	STATUS
41	Network Certificate Inventory	Collect certs across network; dedupe; list SANs & expiries. Sweeps addresses in the netblock collecting every TLS certificate presented, building an inventory.	—	✅ pass
42	SSH Banner & Key Fingerprinter	Grab SSH banners & fingerprints across hosts/ports. Opens SSH sockets with paramiko to collect banners and host key fingerprints.	—	🕒 long-running
43	SNMP Public Community Checker	Test SNMP v2c communities for info leakage. Attempts SNMP reads with the default 'public' community string - a classic default-credential exposure.	—	✅ pass
44	UDP Service Sampler	Lightweight probes to classify common UDP services. Probes common UDP services with protocol-specific payloads and records responders.	—	🕒 long-running
139	SNMP Bulk Walk	Walk the SNMP OID tree of a host using a community string. Performs a full SNMP bulk walk of the OID tree when a community string is accepted.	—	✅ pass
140	NetBIOS Name Query	Query the NetBIOS name service for host and workgroup names. Issues NetBIOS name-service queries to recover Windows host and workgroup names.	—	✅ pass
141	TTL Analysis	Infer operating system and hop distance from IP TTL values. Compares DNS TTLs against observed ICMP TTLs to infer the OS family and hop distance.	—	✅ pass
142	IRR Routing Registry Analyzer	Compare RADB and RIPE route objects for the target ASN. Looks up the prefix in the RIPE IRR database to compare routing intent against reality.	—	⚙️ needs ASN-routable IP
143	Dual Stack Diff	Diff IPv4 versus IPv6 responses served for the same host. Fetches the site over IPv4 and IPv6 and hashes both responses to surface content drift.	—	✅ pass

ID	MODULE	WHAT IT DOES & HOW IT WORKS	KEYS	STATUS
144	DNS CAA Checker	Check CAA records controlling which CAs may issue certificates. Reads CAA records to show which certificate authorities are permitted to issue for the domain.	—	✓ pass
145	Decoy DNS Beacon	Generate decoy DNS lookups to observe resolver behaviour. Plants uniquely-named DNS lookups and watches which resolvers replay them - detects DNS interception.	—	✓ pass
146	Geo IP Spoof Detection	Cross-check geolocation providers for conflicting answers. Cross-checks the IP's geolocation across several providers to spot spoofed or inconsistent geo data.	Ipinfo	✓ pass
147	Passive DNS History	Look up historical passive DNS records for the target. Retrieves historical passive-DNS records from SecurityTrails/HackerTarget to show past IP ownership.	Securitytrails	✓ pass
154	JARM TLS Fingerprint	Active JARM fingerprint of the server's TLS stack (matches public JARM datasets). Sends the ten standard JARM Client Hello probes over raw sockets and fuzzy-hashes the server's cipher/version/extension choices into a 62-character fingerprint.	—	✓ pass

4.2 Web Application Analysis (50 modules)

ID	MODULE	WHAT IT DOES & HOW IT WORKS	KEYS	STATUS
45	Archive History	Retrieve historical site snapshots. Queries the Wayback Machine and Common Crawl indexes for archived URLs of the domain.	—	✓ pass
46	Broken Links Detection	Crawl site & detect broken links. Crawls the page's links and requests each one concurrently, reporting 4xx/5xx targets.	—	✓ pass
47	Carbon Footprint	Estimate environmental impact of page loads. Estimates page energy use and CO2 per visit via the Website Carbon API plus transfer weight.	Website_Carbon	✓ pass
48	CMS Detection	Identify CMS platforms by signature. Matches HTML markers, meta generators, paths and cookies against CMS fingerprints.	—	✓ pass

ID	MODULE	WHAT IT DOES & HOW IT WORKS	KEYS	STATUS
49	Cookies Analyzer	Inspect cookies for security/privacy attributes. Parses Set-Cookie headers for Secure/HttpOnly/SameSite flags and decodes JWT-shaped cookie values.	—	✓ pass
50	Content Discovery	Discover hidden directories/files/endpoints. Asynchronously requests a wordlist of paths to discover unlinked content.	—	✓ pass
51	Crawler	Crawl site & map structure. Walks the site's internal links to build a page inventory.	—	✓ pass
52	Robots.txt Analyzer	Parse robots.txt for hidden/disallowed paths. Fetches robots.txt and highlights disallowed paths, which often name sensitive directories.	—	✓ pass
53	Directory Finder	Scan for common unlisted directories. Brute-forces common directory names with a thread pool and reports the ones that resolve.	—	✓ pass
54	Email Harvesting	Extract emails from site pages. Scrapes rendered HTML and mailto links for email addresses.	—	✓ pass
55	Performance Monitoring	Measure response time & load performance. Runs the page through Google PageSpeed Insights and reports Core Web Vitals.	Google	✓ pass
56	Quality Metrics	Assess site UX/content quality heuristics. Scores page quality signals - markup validity, meta completeness, asset weight.	—	✓ pass
57	Redirect Chain	Follow redirects; analyze safety & loops. Follows the redirect chain hop by hop, printing each Location and status.	—	✓ pass
58	Sitemap Parsing	Parse sitemap.xml; enumerate URLs. Parses sitemap.xml (including sitemap indexes) to enumerate declared URLs.	—	✓ pass
59	Social Media Presence Scan	Identify linked social media profiles. Searches engines and the page itself for links to the brand's social profiles.	—	✓ pass

ID	MODULE	WHAT IT DOES & HOW IT WORKS	KEYS	STATUS
60	Technology Stack Detection	Fingerprint technologies & frameworks in use. Fingerprints frameworks, servers and libraries from headers, HTML markers and script paths.	—	✓ pass
61	Third-Party Integrations	Discover external services integrated into site. Identifies embedded third-party services - analytics, chat, tag managers - from outbound references.	—	✓ pass
62	JavaScript File Analyzer	Extract endpoints & secrets from JS files. Downloads every external script and greps for URLs, emails and rough secret patterns.	—	✓ pass
63	CORS Misconfiguration Scanner	Detect overly permissive CORS settings. Replays requests with forged Origin headers to detect reflective or wildcard CORS trust.	—	✓ pass
64	Login Page Brute Identifier	Locate & fingerprint login/auth pages. Locates login forms and judges whether they expose brute-forceable authentication.	—	✓ pass
65	Hidden Parameter Discovery	Fuzz hidden GET/POST parameters. Probes candidate query parameters and diffs responses to reveal undocumented inputs.	—	✓ pass
66	Clickjacking Test	Check anti-framing headers & behavior. Checks X-Frame-Options and CSP frame-ancestors to determine whether the page can be framed.	—	✓ pass
67	Form Grabber	Enumerate forms & field metadata. Extracts every form, its method, action and input fields - the app's input surface.	—	✓ pass
68	Favicon Hashing	MD5 hash favicon to infer technologies. Downloads the favicon and computes its hash for pivoting to other hosts running the same app.	—	✓ pass
69	HTML Comments Extractor	Parse HTML comments for hidden notes/secrets. Extracts HTML comments, which frequently leak paths, credentials and TODO notes.	—	✓ pass
70	CAPTCHA Presence Checker	Detect CAPTCHA widgets across pages. Detects whether CAPTCHA protection is present on interactive endpoints.	—	✓ pass

ID	MODULE	WHAT IT DOES & HOW IT WORKS	KEYS	STATUS
71	JavaScript Obfuscation Detector	Highlight obfuscated or packed JS. Scores JavaScript for obfuscation markers - packers, hex-encoded strings, eval density.	—	✓ pass
72	Virtual Host Fuzzer	Host header brute to reveal hidden vhosts. Sends varied Host headers against the same IP to discover virtual hosts.	—	✓ pass
73	Session Cookie Lifetime Checker	Measure session cookie longevity. Measures session cookie lifetimes and flags excessively long-lived sessions.	—	✓ pass
74	HTML5 Feature Abuse Detector	Spot risky HTML5 API usage. Looks for risky HTML5 features - postMessage, localStorage, WebRTC - that widen client-side attack surface.	—	✓ pass
75	Autocomplete Vulnerability Checker	Detect sensitive fields with autocomplete enabled. Finds password and sensitive fields left with autocomplete enabled.	—	✓ pass
76	Embedded Object Hunter	Find embedded PDFs/SWF/objects. Hunts embedded objects - Flash, applets, iframes - that indicate legacy or risky content.	—	✓ pass
77	Multi-Language URL Tester	Probe language/locale path handling. Requests locale variants of URLs to map multi-language routing behaviour.	—	✓ pass
78	Pixel Tracker Finder	Detect tracking pixel beacons. Identifies tracking pixels and beacons embedded in the page.	—	✓ pass
79	SEO Abuse Detector	Spot hidden/abusive SEO content. Detects SEO abuse patterns such as cloaking and hidden keyword stuffing.	—	✓ pass
80	Dependency JS/CDN Scanner	Inventory external JS libs & versions. Inventories JavaScript dependencies and the CDNs serving them.	—	✓ pass
81	WebSocket Endpoint Sniffer	Discover ws:// / wss:// endpoints. Discovers WebSocket endpoints referenced in scripts and attempts handshakes.	—	✓ pass

ID	MODULE	WHAT IT DOES & HOW IT WORKS	KEYS	STATUS
82	API Schema Grabber	Attempt to fetch OpenAPI/Swagger schemas. Probes well-known API schema paths - openapi.json, swagger.json - and parses what it finds.	—	✓ pass
83	Lazy-Load Resource Finder	Detect resources loaded dynamically (scroll/JS). Finds lazily-loaded resources that a naive crawler would miss.	—	✓ pass
84	HTTP Method Enumerator	Crawl & test supported HTTP verbs per URL. Sends OPTIONS and unusual verbs to enumerate accepted HTTP methods, flagging PUT/DELETE/TRACE.	—	✓ pass
85	GraphQL Introspection Probe	Discover GraphQL endpoints; attempt schema introspection. Sends a GraphQL introspection query to see whether the schema is publicly readable.	—	✓ pass
86	File Upload Surface Finder	Crawl & detect file upload forms/JS hints. Locates file upload forms and endpoints - a high-value attack surface.	—	✓ pass
87	DOM Sink Scanner	Scan HTML/JS for XSS sinks (eval, innerHTML, etc.). Parses JavaScript for DOM XSS sinks (innerHTML, eval, document.write) and the sources feeding them.	—	✓ pass
88	Cache Behavior Analyzer	Compare caching behavior; detect poisoning risks. Crawls pages and analyses caching headers and Vary behaviour for cache-poisoning exposure.	—	🕒 long-running
89	Cookie Scope Diff Across Subdomains	Aggregate Set-Cookie across crawl; scope & flag analysis. Compares cookie scope and flags across subdomains to find over-broad domain cookies.	—	✓ pass
90	CSP Deep Analyzer	Collect & parse CSP headers; risk scoring. Parses the CSP into directives and grades weaknesses like unsafe-inline, wildcards and missing default-src.	—	✓ pass
91	Third-Party Script Risk Profiler	Inventory external script hosts; categorize & score. Profiles each third-party script by origin, size and privilege to rank supply-chain risk.	—	✓ pass
92	Static Asset Fingerprinter	Hash JS/CSS; extract library versions; flag outdated. Hashes static assets to fingerprint framework versions and detect unexpected changes.	—	✓ pass

ID	MODULE	WHAT IT DOES & HOW IT WORKS	KEYS	STATUS
148	Email Config	Inspect MX, SPF, DKIM and DMARC configuration for the domain. Resolves MX, SPF, DKIM and DMARC plus MTA-STS policy and BIMI records for a full email posture view.	—	✓ pass
157	Subresource Integrity Checker	Check external scripts/styles for SRI hashes and verify the declared hashes actually match. Collects external scripts/stylesheets, then downloads each and recomputes its SHA-256/384/512 to verify the declared integrity hash actually matches.	—	✓ pass

4.3 Security & Threat Intelligence (50 modules)

ID	MODULE	WHAT IT DOES & HOW IT WORKS	KEYS	STATUS
93	Censys Reconnaissance	Enumerate exposed assets via Censys (API). Queries the Censys Platform API for hosts, certificates and services tied to the target.	Censys_Api_Id, Censys, Censys_Api_Secret	✓ pass
94	Certificate Authority Recon	Examine CA issuance & trust relationships. Collects the issuing CA for the endpoint's certificates and profiles issuance patterns.	—	✓ pass
95	Data Leak Detection	Check for public data leaks & sensitive exposures. Generates likely addresses for the domain and checks them against breach-lookup services.	—	✓ pass
96	Exposed Environment Files Checker	Detect exposed .env/config files. Requests common environment-file paths (.env, .env.local) that leak credentials when served.	—	✓ pass
97	Firewall Detection	Identify firewall/WAF presence heuristically. Fingerprints WAFs and firewalls from block-page markers, headers and response behaviour.	—	✓ pass
98	Global Ranking	Retrieve global popularity ranking metrics. Looks the domain up in the Tranco global ranking list.	—	✓ pass
99	HTTP Headers	Extract HTTP response headers. Retrieves and explains every HTTP response header.	—	✓ pass

ID	MODULE	WHAT IT DOES & HOW IT WORKS	KEYS	STATUS
100	HTTP Security Features	Evaluate security headers (HSTS, CSP, etc.). Audits the presence and quality of individual HTTP security features.	—	✓ pass
101	Malware & Phishing Check	Check blocklists for malware/phishing indicators. Checks the domain and URL against VirusTotal and URLhaus malware/phishing feeds.	Virustotal	✓ pass
102	Pastebin Monitoring	Search paste sites for leaked data mentions. Searches paste sites and GitHub gists for the domain appearing in public pastes.	Github	✓ pass
103	Privacy & GDPR Compliance	Basic privacy/GDPR checks (policies, consent). Inspects cookie banners, trackers and policy links for GDPR/privacy compliance signals.	—	✓ pass
104	Security.txt Check	Retrieve & parse security.txt disclosure info. Fetches /.well-known/security.txt and validates its contact fields.	—	✓ pass
105	Shodan Reconnaissance	Query Shodan for exposed services & vulns. Pulls host, port, banner and vulnerability data for the target from Shodan.	Shodan	✓ pass
106	SSL Labs Report	Pull detailed SSL Labs TLS assessment. Submits the host to the Qualys SSL Labs API and reports the returned grade.	—	✓ pass
107	SSL Pinning Check	Check for SSL/TLS pinning indicators. Tests whether the app pins certificates by presenting an unexpected chain.	—	✓ pass
108	Subdomain Enumeration	Discover subdomains via multiple techniques. Pulls subdomains from crt.sh certificate transparency records.	—	✓ pass
109	Subdomain Takeover	Test for dangling DNS entries vulnerable to takeover. Enumerates subdomains and checks each for dangling-CNAME takeover signatures.	—	✓ pass
110	VirusTotal Scan	Lookup reputation & detections in VirusTotal. Submits the domain to VirusTotal and reports multi-engine verdicts.	Virustotal	✓ pass

ID	MODULE	WHAT IT DOES & HOW IT WORKS	KEYS	STATUS
111	CT Log Query	Query certificate transparency logs for issued certs. Queries certificate transparency logs for every certificate issued to the domain.	—	✓ pass
112	Breached Credentials Lookup	Check breach datasets for exposed credentials. Checks credentials against HIBP using the k-anonymity range API, so no full hash leaves the host.	Hibp	✓ pass
113	Cloud Bucket Exposure	Detect open S3/Azure/GCP buckets tied to domain. Generates bucket-name permutations and probes S3, GCS and Azure for public listings.	—	🕒 long-running
114	JWT Token Analyzer	Decode and inspect JWT algorithms & claims. Decodes JWTs found on the site and audits algorithm, expiry and signature weaknesses.	—	✓ pass
115	Exposed API Endpoints	Crawl and list publicly reachable API endpoints. Probes common API paths and documents which respond, including unauthenticated ones.	—	✓ pass
116	Git Repository Exposure Check	Detect exposed .git directories and artifacts. Requests .git/HEAD and related paths to detect an exposed repository.	—	✓ pass
117	Typosquat Domain Checker	Generate and check typo variants for malicious domains. Generates typo variants of the domain and resolves them to find squatters.	—	✓ pass
118	SPF / DKIM / DMARC Validator	Assess email auth posture and alignment. Validates SPF, DKIM and DMARC records and reports their tags.	—	✓ pass
119	Open Redirect Finder	Probe redirect parameters for open redirect vulnerabilities. Injects external URLs into redirect-shaped parameters and watches for off-site 3xx responses.	—	🕒 long-running
120	Rate-Limit & WAF Bypass Test	Probe throttling and WAF bypass behaviors. Sends bursts of requests with varied headers to measure rate limiting and probe WAF bypasses.	—	🕒 long-running
121	Security Changelog Diff	Compare security header/config changes over time. Snapshots security headers to a baseline file and diffs future runs against it.	—	✓ pass

ID	MODULE	WHAT IT DOES & HOW IT WORKS	KEYS	STATUS
122	Session Hijacking (Passive)	Analyze cookie/session flags for hijacking risk. Passively inspects session cookies and tokens for hijacking weaknesses - no active session attack.	—	✓ pass
123	Rogue Certificate Check	Detect suspicious or duplicate certificates. Compares certificates seen in CT logs against what the server presents to spot rogue issuance.	—	✓ pass
124	JS Malware Scanner	Heuristic scan of JavaScript for malware indicators. Scans served JavaScript for malware and skimmer signatures.	—	✓ pass
125	Cloud Service Enumeration	Detect exposed cloud/devops services (Jira, Jenkins, etc.). Detects which cloud services and SaaS providers the domain depends on.	—	✓ pass
126	Rogue Subdomain Resolver	Monitor for newly resolving previously dead subdomains. Resolves candidate subdomains to find records pointing at infrastructure the owner no longer controls.	—	✓ pass
127	Bug Bounty Program Finder	Identify bug bounty/disclosure program links. Checks whether the domain runs a public bug bounty or VDP programme.	—	✓ pass
128	Custom Wordlist Generator	Build tailored recon wordlists (paths, usernames, emails). Builds a target-specific wordlist from the site's own content for later fuzzing.	—	✓ pass
129	Threat Feed Correlator	Aggregate multi-feed reputation & threat intelligence. Correlates the target against AbuseIPDB, OTX and VirusTotal threat feeds.	Abuseipdb, Virustotal	✓ pass
130	Attack Surface Delta	Diff two Perimetry reports; highlight adds/removals. Asynchronously gathers subdomains, resolves them and scans ports to diff the attack surface over time.	—	🕒 long-running
131	Passive CVE Mapper	Map discovered product/version hints to NVD CVEs. Maps passively-observed software versions to CVE records from the NVD feed.	—	✓ pass
132	Security Contact Gap Finder	Collect security contacts from security.txt, WHOIS, site. Determines whether a security contact exists across security.txt, RDAP and DNS.	—	✓ pass

ID	MODULE	WHAT IT DOES & HOW IT WORKS	KEYS	STATUS
133	Domain Shadowing Detector	CT + passive DNS to spot high-entropy subdomain bursts. Looks for attacker-created subdomains under a legitimate domain using CT and passive DNS sources.	—	✓ pass
134	IP Reputation Trending	Compare AbuseIPDB & VT metrics across time windows. Tracks the IP's reputation scores over time across AbuseIPDB and VirusTotal.	Abuseipdb, Virustotal	✓ pass
149	Dark Web Monitoring	Search public breach and paste sources for the domain. Queries paste and breach-index services for mentions of the domain.	—	✓ pass
150	IP Reputation Check	Check the resolved IP against reputation blocklists. Checks the resolved IP against AbuseIPDB and IPQualityScore blocklists.	Abuseipdb, Ipqualityscore	✓ pass
151	TLS Security Config	Audit TLS protocol versions and configuration weaknesses. Audits negotiated TLS versions, cipher strength, forward secrecy and known protocol weaknesses.	—	✓ pass
152	JS Secret / API Key Scanner	Scan inline and external JavaScript for leaked API keys, tokens and private keys. Downloads inline and external JavaScript, matches 22 vendor credential patterns plus a Shannon-entropy filter, and reports each hit masked with a severity grade.	—	✓ pass
153	Security Headers Grade	Score HTTP security headers into an A-F grade and flag info-disclosure headers. Reads the response headers and scores seven controls by weight into an A+ to F grade, deducting for information-disclosure headers.	—	✓ pass
155	CVE Enrichment from Tech Stack	Fingerprint tech stack versions and look up known CVEs via the NVD feed. Fingerprints product versions from headers, meta generators and script paths, then queries the NVD 2.0 API per component and ranks results by CVSS.	Nvd	✓ pass
156	Subdomain Takeover (Deep)	Enumerate subdomains from CT logs and flag dangling CNAMEs against a takeover fingerprint DB. Merges crt.sh and CertSpotter subdomain sets, resolves each CNAME, and matches it against a 20-service takeover fingerprint database before confirming with a response marker.	—	✓ pass

ID	MODULE	WHAT IT DOES & HOW IT WORKS	KEYS	STATUS
158	Document Metadata Extractor	Harvest author names, software versions and internal paths from public PDF and Office documents. Discovers linked PDF/Office files (homepage, sitemap, one crawl hop), fetches head plus a Range-requested tail, then parses the PDF info dictionary, the XMP packet and OOXML docProps.	—	✓ pass

5. Review status & known issues

5.1 What the run proved

- **No module crashes.** Across 154 live executions there was not a single Python traceback - unusual for a suite this size.
- **Catalog integrity is clean:** IDs unique, names unique, every catalog entry resolves to a real file, no orphaned module files.
- **All 188 Python files compile,** and every core/CLI/utility module imports.

5.2 Long-running modules

Eight modules exceeded the 60-second test cap. None are broken - they are doing real work and were mid-progress when cut off:

- 9 Open Ports Scan
- 42 SSH Banner & Key Fingerprinter
- 44 UDP Service Sampler
- 88 Cache Behavior Analyzer
- 113 Cloud Bucket Exposure
- 119 Open Redirect Finder
- 120 Rate-Limit & WAF Bypass Test
- 130 Attack Surface Delta

Give them a longer `timeout`, or expect `runall` to spend most of its wall-clock here.

5.3 API keys the CLI cannot manage

The `api` command manages 7 services, but modules reference **8 further keys** with no way to set them from the CLI - they must be added to `.env` by hand:

KEY	USED BY
ABUSEIPDB_API_KEY	129, 134, 150

KEY	USED BY
SECURITYTRAILS_API_KEY	147
OTX_API_KEY	6
IPINFO_API_KEY	146
IPQUALITYSCORE_API_KEY	150
NVD_API_KEY	155
GITHUB_TOKEN	102
WEBSITE_CARBON_API_KEY	47

SSL_LABS_API_KEY is the mirror image: declared in settings but used by no module.

5.4 Legacy modules

Five modules execute at import time because they lack an `if __name__ == "__main__":` guard - `redirect_chain`, `server_info`, `server_location`, `ssl_chain`, `subdomain_enum`. Harmless today, since the runner always launches them as subprocesses, but they cannot be imported or unit-tested.

5.5 No automated test suite

`make test` and `make lint` reference `tests/` and `docs/` directories that do not exist. For a project of this size, that is the single largest structural gap.

6. Where this sits as a security project

Strengths. Breadth is genuinely unusual - 154 modules spanning network, web and threat intelligence, in one consistent CLI with favourites, profiles, per-module options and automatic reporting. The process-isolation design means the suite degrades gracefully instead of collapsing. Several modules (JARM, SRI verification, HIBP k-anonymity, RPKI) are implemented to a standard matching dedicated tools.

The gap to "professional". Three things separate Perimetry from a tool a team would standardise on:

1. **No test suite.** Nothing prevents a regression in module 47 from shipping.
2. **No machine-readable output.** Modules print `rich` tables; severity is recovered by `*regex-matching the printed text*`. A JSON result schema would make Perimetry scriptable and CI-integrable - the highest-leverage change available.

3. **Depth over breadth from here.** The next ten modules matter less than making the existing weak ones (§3) as good as the flagship ten.

Recommended next steps, in order:

1. Add a JSON output mode with a shared result schema.
2. Add `tests/` with smoke tests that assert each module exits 0 and prints its completion marker - the sweep used for this document is effectively that test, and it should be committed.
3. Extend the `api` command to cover the 8 unmanaged keys (§5.3).
4. Rebuild or retire the weakest modules (§3) rather than adding new ones.
5. Normalise the five legacy modules onto `run(target, threads, opts)` .

Generated from a full execution run of all 154 modules plus static analysis of the codebase. Status values are empirical, not aspirational.